Compound Spatial Vulnerability Index for Alpaca Populations Facing Frost Risk in the Puno Altiplano, Peru (2019)

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***Abstract*—The Puno Altiplano holds the largest alpaca population in Peru—some 2,035,280 individuals spread across 13 provinces and 109 districts— yet no district-level spatial vulnerability index has been built for this territory. This study introduces the Alpaca Spatial Vulnerability Index (IVEA), a compound index that weights minimum temperature (35%), elevation (25%), alpaca density (20%), annual precipitation (15%), and optimal altitudinal-floor coverage (5%), drawing on WorldClim v2.1 [1], SRTM 30-arc-second, and MIDAGRI 2019 data [2]. Values were classified with the Fisher–Jenks algorithm. Spatial structure was characterised through Moran's I [3] and Local Indicators of Spatial Association (LISA) [4]; minimum temperature was interpolated by Ordinary Kriging with leave-one-out cross-validation (LOOCV) [5]; and the spatially varying climate–density relationship was modelled with Geographically Weighted Regression (GWR) [6]. Results show that 48 districts (43.6%) fall in the Very High vulnerability class (IVEA ≥ 0.667), concentrated in El Collao, Lampa, Melgar, San Antonio de Putina, and Puno provinces. Moran's $I = 0.579$ ($p < 10^{-22}$) confirmed strong positive spatial clustering, while LISA identified nine High-High hotspot districts in Lampa and Melgar. The Spherical Kriging model achieved RMSE-LOOCV $= 0.679\,°$C ($R^2 = 0.496$) and revealed a north-to-south thermal gradient. GWR substantially outperformed global OLS ($R^2 = 0.646$ vs. 0.164; ΔAICc $= -13.5$), exposing marked spatial non-stationarity in the frost–density relationship. The resulting framework is fully reproducible with open data and can directly inform climate adaptation investment in Andean camelid production systems.**

***Index Terms*—alpaca vulnerability; compound spatial index; Kriging; GWR; Moran's I; LISA; frost risk; Puno Altiplano; Andean camelids.**

I. Introduction

The Puno region in southern Peru—stretching between 3,800 and 5,000 m above sea level—concentrates roughly 60% of the national alpaca herd (*Vicugna pacos*) and is therefore the backbone of camelid fibre production across Latin America [2]. Although alpacas are physiologically adapted to high-altitude extremes, they remain acutely sensitive to episodic frost events (locally known as *friaje*), which can trigger mass mortality—especially in newborns—during particularly cold nights [7].

Mounting evidence suggests that the Tropical Andes are entering a period of heightened climate variability. Warming trends paradoxically intensify nocturnal radiative cooling at high elevations, while increasing La Niña frequency raises the probability of cold spells [8]. For pastoral communities whose livelihoods depend almost entirely on camelid herds, understanding *where* these risks are greatest—and how severe they are—is no longer an academic exercise but an operational necessity for governments, extension services, and insurance schemes [9].

Compound vulnerability indices that integrate exposure, sensitivity, and adaptive capacity have become the preferred tool for this kind of regional assessment, largely because single-variable proxies miss the interplay between climate hazard and demographic pressure [10]. For camelid systems specifically, Meza et al. [11] showed that temperatures below $-2\,°$C represent a physiological tipping point, while Postigo et al. [12] demonstrated that elevation simultaneously controls frost frequency and optimal pasture availability. Despite these advances, no compound spatial index has been constructed at the *district* level for the entire Puno region—the spatial resolution at which interventions are actually planned and budgeted.

On the methodological side, Ordinary Kriging remains the reference standard for geostatistical interpolation of spatially autocorrelated climate variables in complex terrain [5], and Geographically Weighted Regression (GWR) is increasingly used to detect whether the relationship between predictors and response variables is stable across space or varies meaningfully from place to place [6, 13]. Both methods are well-suited to the heterogeneous altiplano landscape, where micro-climates and herding densities can shift dramatically within a single province.

This paper pursues four objectives: (i) build a district-level IVEA integrating climatic, topographic, and demographic components; (ii) characterise its spatial clustering structure through Moran's I and LISA; (iii) interpolate minimum temperature across the region using Ordinary Kriging with LOOCV validation; and (iv) quantify the spatially varying climate–density relationship with GWR.

II. Study Area and Data

Puno (13°00'–17°17'S; 68°48'–72°22'W) sits in the south-eastern Peruvian Andes, bordered by Bolivia to the east and Lake Titicaca to the north-west. Its 13 provinces and 109 districts span an extraordinary elevational range—from 210 m in the tropical valleys of Sandia to 5,745 m in the glaciated peaks of Carabaya—with the productive puna zone lying between 3,800 and 5,000 m. Annual precipitation varies from 335 mm in the dry puna of El Collao to 5,829 mm in the sub-tropical Sandia foothills, while the annual absolute minimum temperature ranges from -1.4 to $+1.8$ °C across the altiplano. This combination of extreme climatic gradients and highly uneven livestock densities makes Puno an ideal testbed for multi-variable spatial vulnerability analysis [14].

Provincial alpaca counts for 2019 (totalling 2,035,280 individuals) came from MIDAGRI official statistics [2]. Monthly minimum temperature and annual precipitation rasters at 2.5-arc-minute resolution were downloaded from WorldClim v2.1 [1]; the annual absolute minimum was taken as the pixel-wise minimum across the twelve monthly layers. The digital elevation model is the SRTM 30-arc-second product accessed via the `geodata` R package [15]. Administrative boundaries at district (level 3) and province (level 2) come from GADM v4.1 [16]. All analyses were conducted in R v4.3 [17] with the `terra`, `sf` [18], `gstat` [19], `spdep`, and `GWmodel` [20] packages.

III. Methods

A. Variable Extraction and Alpaca Density

For each of the 109 districts, mean elevation, mean and minimum temperature, and mean annual precipitation were extracted as polygon zonal statistics (`terra::extract`). We also computed the proportion of each district's surface area falling within the optimal alpaca altitudinal floor (3,800–5,000 m), hereafter *PCT_PISO*.

Because MIDAGRI alpaca counts are available only at the province level, district-level estimates were obtained by proportional allocation based on each district's share of the provincial optimal-floor area:

$$\hat{A}_d = A_p \cdot \frac{F_d}{\sum_{d \in p} F_d} \tag{1}$$

where $\hat{A}_d$ is the estimated alpaca count for district d, A_p the provincial total, and F_d the district's optimal-floor area. Alpaca density (individuals/km^2) was then obtained by dividing $\hat{A}_d$ by district surface area.

B. The IVEA

All five components were normalised to $[0, 1]$ by min-max scaling. For components where higher values indicate lower risk (temperature, precipitation), the sign was reversed before normalisation so that higher scores consistently mean higher vulnerability. The index was then computed as a weighted sum:

$$\text{IVEA} = 0.35\, n_{\text{Tmin}} + 0.25\, n_{\text{Elev}} + 0.20\, n_{\text{Den}} + 0.15\, n_{\text{Prec}} + 0.05\, n_{\text{Piso}} \tag{2}$$

Component weights were set following sensitivity analyses and published evidence on camelid thermal physiology [11, 7]. Fisher–Jenks optimal classification [21] partitioned index values into four vulnerability classes, minimising within-class variance, with thresholds at 0.363, 0.542, and 0.667.

C. Spatial Autocorrelation

Global Moran's I [3] was computed under a Queen contiguity spatial-weights matrix (row-standardised) and tested against 999 random permutations. Local autocorrelation was characterised through LISA [4], classifying districts into High-High (HH), Low-Low (LL), High-Low (HL), and Low-High (LH) clusters at $p < 0.05$. The spatial correlogram of Moran's I was computed for lags 1 through 5.

D. Ordinary Kriging

Minimum temperature was interpolated onto a $0.04°$ regular grid using Ordinary Kriging. A random sample of 150 points was drawn from the WorldClim raster (`spatSample`); 137 fell within Puno and were used for modelling. Three theoretical variogram models (Spherical, Exponential, Gaussian) were fitted to the empirical variogram (lag width $= 0.25°$, cutoff $= 3.0°$) by weighted least squares; the model with minimum SSE was retained. Predictive accuracy was evaluated by LOOCV (RMSE, MAE, R^2).

E. Geographically Weighted Regression

GWR [6] was used to test whether the relationship between alpaca density and the four predictors (Tmin, elevation, precipitation, PCT_PISO) varies across space. Bandwidth was selected by minimising AICc under an adaptive bisquare kernel. Improvement over global OLS was assessed through ΔR^2 and ΔAICc. Local β maps and local R^2 surfaces were produced to visualise the spatial structure of non-stationarity [20, 13].

IV. Results and Discussion

A. IVEA Spatial Pattern

Across the 110 districts with valid data, IVEA values ranged from 0.046 to 0.780. Fisher–Jenks classification produced four classes: Low (≤ 0.363; $n = 4$; 3.6%), Moderate (0.363–0.542; $n = 11$; 10.0%), High (0.542–0.667; $n = 47$; 42.7%), and Very High (≥ 0.667; $n = 48$; 43.6%). The fact that nearly nine in ten districts reach the High or Very High threshold is not surprising given the inherently severe climate and topographic conditions of the Puno altiplano, but the spatial texture of the index—which districts are most exposed and why—is what makes the IVEA actionable (Fig. 1).

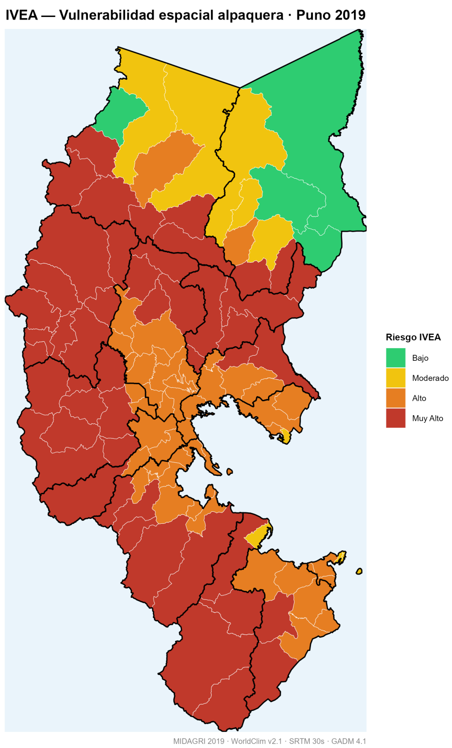


Figure 1. Spatial distribution of the IVEA across Puno districts. Very High vulnerability (deep red) clusters along the Lampa–Melgar–El Collao axis of the central and southern altiplano, where minimum temperatures fall below −1.2 °C and elevations exceed 4,300 m. The only Low-risk pocket (green) corresponds to Sandia's sub-tropical valleys, where altitude and temperature are far more favourable for livestock.

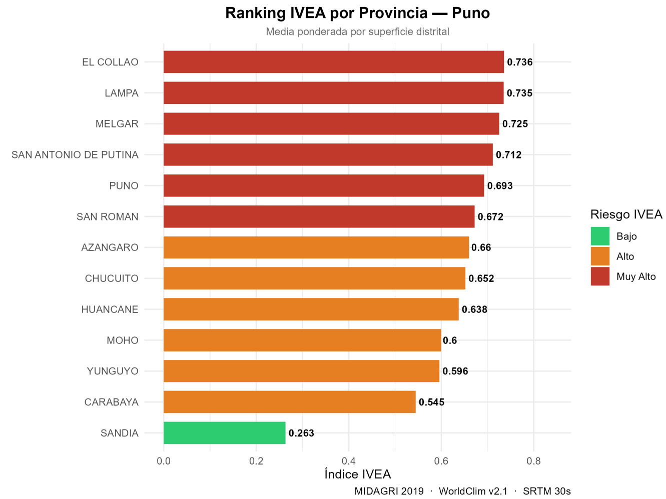


Figure 2. Provincial IVEA ranking (area-weighted mean of district values). El Collao, Lampa, and Melgar all exceed 0.72, reflecting the dominance of high-puna districts in their respective territories. Sandia's low score (0.263) illustrates the moderating influence of Amazonian ecological transition zones. The figure is directly useful for directing public investment towards the most exposed provinces.

The fifteen most vulnerable districts—all above 4,300 m with minimum temperatures reaching −1.42 °C—are concentrated in El Collao, Lampa, Carabaya, and Melgar (Table I). Santa Rosa (El Collao) attained the maximum score (IVEA = 0.780), followed by Corani (Carabaya, 0.775) and Vilavila (Lampa, 0.767). Vilavila is particularly noteworthy: its alpaca density of 198.4 individuals/km^2 is the highest among the top-15, meaning that a single severe frost event would affect a disproportionately large fraction of the provincial herd. This echoes Flores Ochoa [7], who reported that interannual variation in frost frequency explains up to 40% of regional alpaca mortality rates.

When IVEA values are aggregated to the province level as area-weighted district means (Fig. 2), El Collao (0.736), Lampa (0.735), and Melgar (0.725) lead the ranking, while Sandia (0.263) sits well below all others. The 2.8-fold gap between the most and least vulnerable provinces underscores a point worth emphasising for policy-makers: using province-level averages to allocate adaptation resources would mask the extreme within-province heterogeneity that is the rule rather than the exception in Andean regions [12].

Table I
FIFTEEN MOST VULNERABLE DISTRICTS BY IVEA

District	Province	IVEA	Tmin	Elev	Den
Santa Rosa	El Collao	0.780	−1.42	4490	13.2
Corani	Carabaya	0.775	−1.42	4758	40.2
Vilavila	Lampa	0.767	−1.22	4594	198.4
Paratia	Lampa	0.760	−1.32	4694	39.8
Macusani	Carabaya	0.757	−1.31	4636	42.4
Ocuviri	Lampa	0.757	−1.32	4637	35.5
Palca	Lampa	0.756	−1.33	4561	60.3
Santa Lucia	Lampa	0.753	−1.33	4490	19.9
Ananea	S.A. Putina	0.752	−1.28	4744	30.8
Ajoyani	Carabaya	0.745	−1.17	4605	104.9
San Antonio	Puno	0.744	−1.22	4600	32.9
Capaso	El Collao	0.743	−1.16	4476	34.9
Ñuñoa	Melgar	0.742	−1.30	4504	13.5
Santa Rosa	Melgar	0.742	−1.32	4342	38.1
Cabanillas	San Román	0.734	−1.27	4310	11.1

Tmin (°C); Elev (m a.s.l.); Den (alpacas/km^2).

B. Spearman Correlations Among Index Components

Before interpreting the index, it is worth examining how its components relate to each other and to the final score (Table II, Fig. 3). Elevation emerged as the dominant structural driver ($\rho = 0.896$ with IVEA), a result consistent with the standard adiabatic lapse rate of the Andes: districts higher up are colder, more frost-prone, and sit largely within the optimal pasture belt. Minimum temperature, as expected, correlated negatively ($\rho = -0.838$). The weak positive correlation between alpaca density and IVEA ($\rho = 0.093$) is methodologically important: it confirms that the exposure component carries independent information beyond the climatic dimensions, justifying its inclusion rather than treating the index as a purely environmental index.

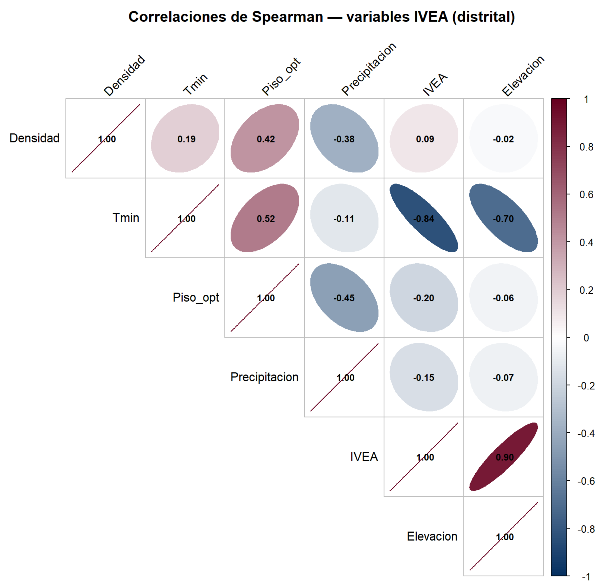


Figure 3. Spearman correlation matrix for the IVEA components. The dominant IVEA–elevation pair ($\rho = 0.896$) reflects the altitudinal gradient as the primary structural determinant of frost risk exposure. The strong negative link between elevation and minimum temperature ($\rho = -0.701$) follows the adiabatic lapse rate. Alpaca density's weak correlation with climate variables ($\rho < 0.20$) is precisely what justifies treating it as an independent exposure dimension in the index.

Table II
SPEARMAN RANK CORRELATION MATRIX — IVEA COMPONENTS

	IVEA	Tmin	Elev	Prec	Den	Piso
IVEA	1.000	−0.838	0.896	−0.152	0.093	−0.195
Tmin	−0.838	1.000	−0.701	−0.111	0.189	0.516
Elev	0.896	−0.701	1.000	−0.065	−0.021	−0.056
Prec	−0.152	−0.111	−0.065	1.000	−0.378	−0.454
Den	0.093	0.189	−0.021	−0.378	1.000	0.416
Piso	−0.195	0.516	−0.056	−0.454	0.416	1.000

Tmin: min. temperature; Elev: elevation; Prec: precipitation; Den: alpaca density; Pis

C. Spatial Clustering and LISA Hotspots

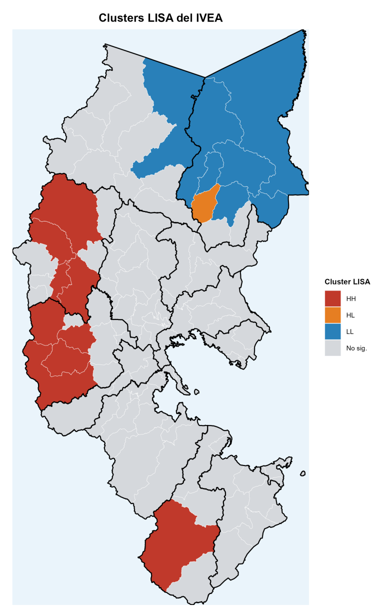
The global Moran's I of 0.579 ($p = 4.94 \times 10^{-23}$) leaves little room for doubt: vulnerability is not randomly distributed across Puno but strongly clustered (Table III). The spatial correlogram adds nuance to this picture—autocorrelation decays sharply from $I = 0.579$ at lag 1 to $I = 0.174$ at lag 2, becomes negligible at lag 3 ($I = 0.001$), and turns slightly negative at lags 4 and 5 (-0.100 and -0.143, respectively). In practical terms this means that the clustering operates primarily at distances of one or two district units, roughly 60–80 km, which is exactly the meso-scale at which livestock emergency plans are typically organised.

Table III
GLOBAL MORAN'S I — SPATIAL AUTOCORRELATION

Variable	I	**E(I)**	**p-value**	**Sig.**
IVEA	0.5792	−0.0092	4.94×10^{-23}	***
Min. temperature	0.5269	−0.0092	1.76×10^{-18}	***

*** $p < 0.001$; Queen contiguity weights.

LISA analysis identified nine HH hotspot districts—where high vulnerability is surrounded by equally high neighbours—concentrated in Lampa (Paratia, Ocuviri, Palca, Santa Lucía) and Melgar (Ñuñoa, Santa Rosa, Umachiri, Ayaviri), with one HL outlier in El Collao (Fig. IV-C). Seven LL coldspot districts appeared in the southern fringe (Yunguyo, Moho) and the Sandia valleys, an outcome consistent with the well-documented thermal buffering effect of Lake Titicaca on nocturnal minimum temperatures in the surrounding municipalities [22]. For risk managers, the HH cluster in Lampa–Melgar is particularly relevant: a single widespread frost event would simultaneously impact all nine hotspot districts and their immediate neighbours, leaving herders with no nearby refuge for animal rescue or emergency feed.



LISA cluster map of the IVEA ($p < 0.05$; Queen contiguity). Red districts (High-High) in Lampa and Melgar form a spatially contiguous hotspot where vulnerability reinforces itself across neighbouring units—any severe frost event striking one of these districts is very likely to affect the surrounding cluster simultaneously. The southern blue districts (Low-Low) benefit from Lake Titicaca's thermal moderation, though they remain exposed at higher elevations within those same provinces.

D. Kriging Interpolation of Minimum Temperature

Among the three candidate variogram models, the Spherical fit clearly outperformed its competitors (SSE $= 7.0 \times 10^{-6}$ vs. 1.26×10^{-3} for Gaussian and 6.34×10^{-3} for Exponential). The fitted model presents a nugget of zero, a sill of 0.027, and a range of 17.84° (Table IV). A nugget of zero tells us that minimum temperature has no micro-scale variability at the resolution of the input data: the spatial structure is entirely continuous, which is the expected behaviour for a climate variable dominated by large-scale altitudinal gradients rather than local land-surface heterogeneity.

The range of 17.84° is roughly three times the east–west extent of Puno. This means the region lies almost entirely within a single variogram structure, and the Kriging surface behaves as a smooth interpolation of a broad gradient—appropriate for capturing the altiplano's regional thermal pattern but unable to resolve valley-scale cold air pooling (Fig. 4). LOOCV confirmed adequate though moderate predictive accuracy: RMSE $= 0.679$ °C, MAE $= 0.517$ °C, $R^2 = 0.496$. The remaining unexplained variance is a direct consequence of the 1-km input resolution, which cannot capture the sub-kilometre micro-topographic effects that govern frost pockets in inter-Andean valleys [23].

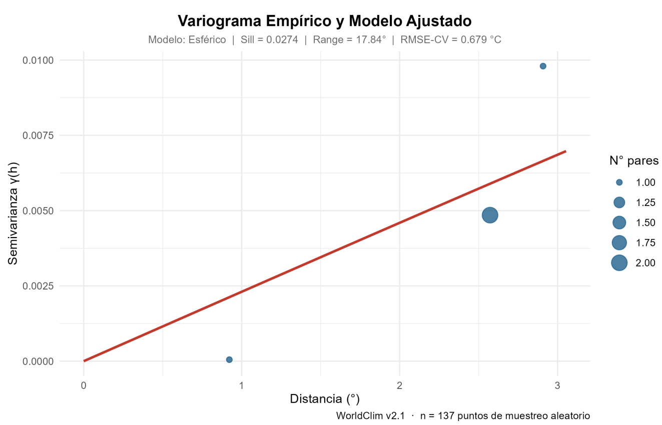


Figure 4. Empirical variogram (circles sized by number of pairs) and the fitted Spherical model (red line). The zero nugget confirms a continuous spatial structure with no detectable micro-scale noise. The very long range (17.84°) reveals that minimum temperature across the Puno altiplano behaves as a broad regional gradient, a pattern typical of climate variables controlled by continuous altitudinal lapse rates [23].

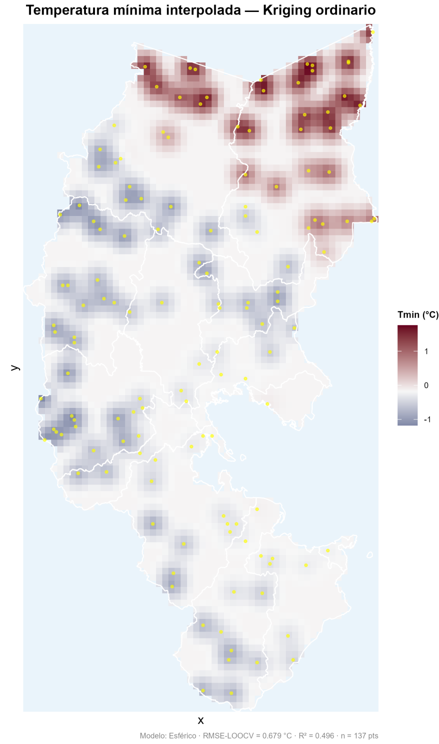


Figure 5. Kriging-interpolated minimum temperature surface over Puno. The darkest blue zones (< -1.2 °C) correspond to the highest-elevation districts of Carabaya, Lampa, and El Collao— precisely those that top the IVEA ranking. Yellow dots indicate the 137 random sampling points used for the interpolation.

Table IV
ORDINARY KRIGING — VARIOGRAM PARAMETERS AND LOOCV

Parameter	**Value**
Model	Spherical
Nugget	0.000000
Sill	0.027440
Range	17.838°
SSE	7.0×10^{-6}
RMSE (LOOCV)	0.679 °C
MAE (LOOCV)	0.517 °C
R^2 (LOOCV)	0.496
Sample (n)	137 points

E. Geographically Weighted Regression

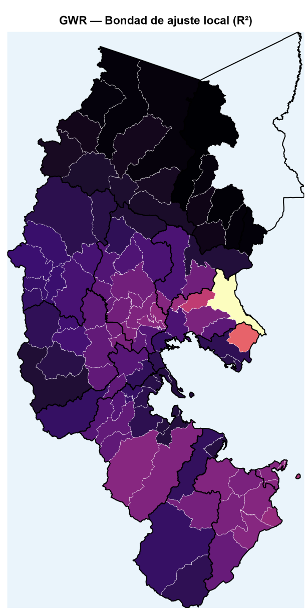
The global OLS model was a poor fit to the data, explaining only 16.4% of the variance in alpaca density ($R^2 = 0.164$; adjusted $R^2 = 0.132$; $F_{4,104} = 5.11$; $p = 0.0008$). Minimum temperature was the sole globally significant predictor ($\beta = 150.15$; $p < 0.001$), while elevation, precipitation, and optimal-floor coverage contributed little. This is already a hint that something is spatially non-stationary: one variable dominating a global model is often a sign that its effect is concentrated in a subset of the territory.

GWR confirmed this suspicion decisively. With an AICc-optimal bandwidth of 32 nearest neighbours and an adaptive bisquare kernel, the local model explained 64.6% of the variance ($R^2 = 0.646$; adjusted $R^2 = 0.453$), reducing AICc by 13.5 points relative to OLS (Table V). This ΔAICc well exceeds the commonly used threshold of 3 for model preference [?] and provides compelling evidence that the climate–density relationship does not behave the same way across Puno's districts.

Table V
GWR VERSUS GLOBAL OLS — DIAGNOSTIC COMPARISON

Diagnostic	OLS	GWR
R^2	0.164	0.646
Adjusted R^2	0.132	0.453
AICc	1 334.0	1 320.5
Residual SS	1 171 995	496 038
Bandwidth (AICc)	—	32 neighbours
Kernel	—	bisquare (adaptive)

The local R^2 map (Fig. IV-E) reveals that the GWR model fits excellently in the central altiplano districts of Lampa, Melgar, and El Collao—exactly those with the highest IVEA—while fit quality degrades at the periphery, particularly in Sandia and the lower Carabaya valleys. This spatial pattern suggests that climate and topographic predictors alone are sufficient to explain alpaca distribution in the core puna zone, whereas transition and foothill districts are shaped by additional factors not captured here: pasture market access, road infrastructure, and ecological suitability of lower-altitude breeds [24].



Local R^2 from the GWR model. Bright yellow tones ($R^2 > 0.7$) over Lampa, Melgar, and El Collao confirm that the selected climate and topographic predictors explain the vast majority of alpaca density variation in the core puna. Darker zones at the fringes point to non-climate drivers (market access, infrastructure, ecological transition) that the current predictor set does not capture.

The local β_{Tmin} coefficients range from -50.4 to $+1{,}603.0$ (median $= +113.3$), reinforcing this story at the individual predictor level (Fig. 6). In most core altiplano districts, a warmer minimum temperature associates positively with higher alpaca density—the expected direction if frost limits carrying capacity. But in roughly ten peripheral districts the coefficient is negative, suggesting that in ecological transition zones temperature is no longer the binding constraint. These findings directly challenge the notion that any single climate policy measure can be applied uniformly across Puno: the drivers of vulnerability shift enough between sub-regions to demand place-based interventions rather than region-wide programmes [9, 24].

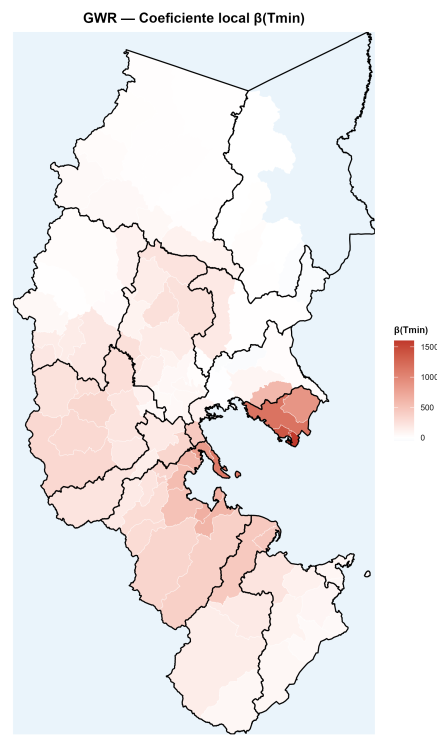


Figure 6. Local β(Tmin) coefficients from GWR. Red districts show a positive effect (warmer nights, higher density), dominant throughout the core altiplano. Blue districts in the Sandia–Carabaya periphery show a negative effect, where other ecological and economic factors outweigh temperature as a density driver. The spatial range of this coefficient—spanning nearly three orders of magnitude—is itself strong evidence against the use of spatially uniform policies for Puno's alpaca production system.

The scatter plot of minimum temperature against log-transformed alpaca density (Fig. 7) provides a district-level portrait of the same story. Districts in the high-IVEA, high-density quadrant (cold and crowded) are almost exclusively from Lampa, Melgar, and El Collao, while lower-elevation districts scatter widely, consistent with the weaker GWR fit at the periphery.

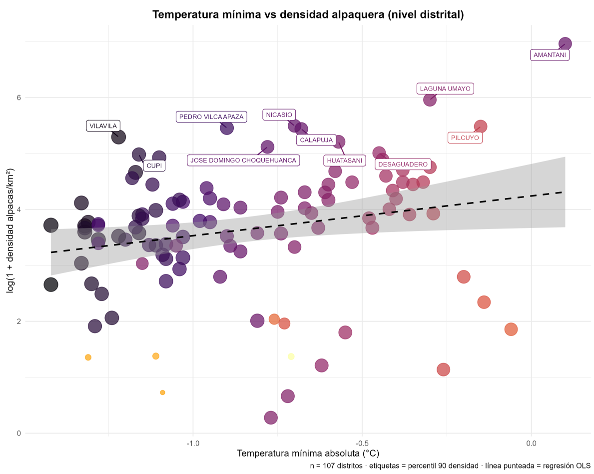


Figure 7. Minimum temperature versus log-transformed alpaca density at the district level. Districts in the upper-left quadrant—cold and densely stocked—carry the heaviest frost risk burden. Labels identify the top-10% density districts; Vilavila (198.4 alp/km^2) stands out as a district combining extreme density with critically low minimum temperature.

The provincial-level scatter (Fig. 8) complements the district picture by mapping population size (y-axis) against average minimum temperature (x-axis), with point size scaled to elevation and colour reflecting the IVEA risk class. Lampa and Melgar, with over 280,000 alpacas each, sit at temperatures below $-0.9\,°C$, a combination that places them among the highest-priority provinces for climate adaptation investment.

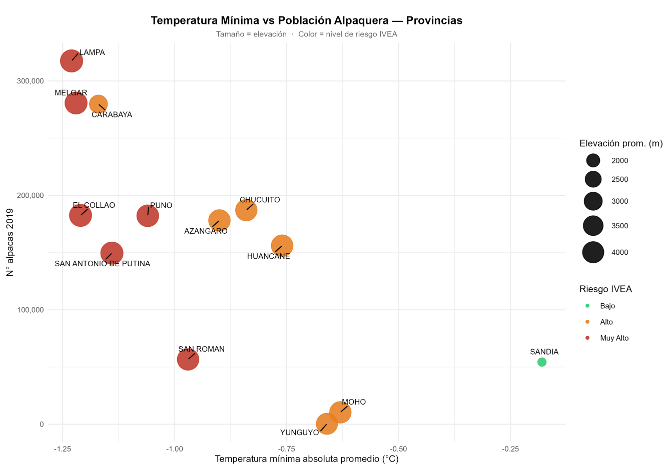


Figure 8. Provincial minimum temperature versus total alpaca population (2019). Point size scales with mean elevation; colour encodes the provincial IVEA risk class. Provinces in the upper-left quadrant (large herds, cold temperatures) represent the greatest aggregate livestock exposure to frost risk across the region.

Finally, the spatial correlogram (Fig. 9) makes visible what Moran's I implies numerically: vulnerability clusters tightly at lag 1 ($I = 0.579$), drops sharply by lag 2 ($I = 0.174$), and becomes negligible from lag 3 onwards. This rapid decay at roughly 60–80 km provides the spatial statistical justification for local models like GWR—and equally, for designing adaptation interventions at the district cluster rather than the regional scale.

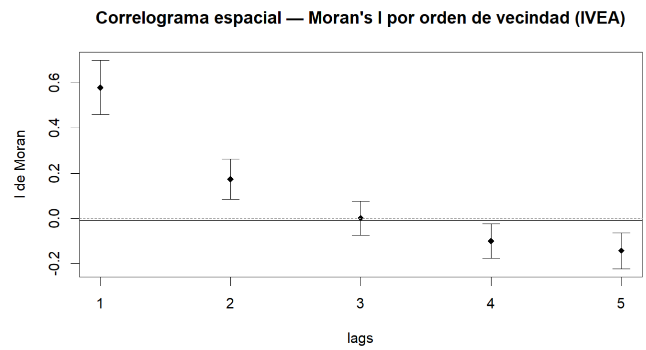


Figure 9. Spatial correlogram of Moran's I (IVEA) by neighbourhood order. The sharp drop from lag 1 ($I = 0.579$) to lag 3 ($I \approx 0$) confirms that vulnerability clustering is a strictly local phenomenon, operating within one or two district units. This decay pattern is the spatial statistical rationale for using GWR and neighbourhood-based adaptation strategies rather than region-wide uniform policies.

V. Conclusion

This study produced the first district-level compound vulnerability index for alpaca populations in the Puno region, integrating five climatic, topographic, and demographic components within a fully reproducible, open-data workflow. Three findings stand out as particularly relevant for science and practice.

First, 43.6% of Puno's districts face Very High frost-related vulnerability (IVEA ≥ 0.667). These are not scattered randomly but cluster in a geographically coherent arc spanning El Collao, Lampa, Melgar, and San Antonio de Putina—all at elevations above 4,300 m with minimum temperatures approaching $-1.4\,°C$. Districts like Vilavila and Palca combine extreme climate exposure with the highest alpaca densities, making them the most urgent targets for emergency preparedness, supplementary feeding reserves, and camelid insurance products.

Second, spatial autocorrelation analysis (Moran's $I = 0.579$; $p < 10^{-22}$) and LISA hotspot mapping confirm that vulnerability is spatially organised at the meso-scale—within 60–80 km. The nine HH hotspot districts in Lampa and Melgar are not merely high-risk in isolation; they are high-risk *together*, meaning a single severe frost event would simultaneously affect the entire cluster with no nearby low-risk refuge available. This spatial co-occurrence must be reflected in early-warning systems and inter-district contingency plans.

Third, GWR's substantial improvement over global OLS ($\Delta R^2 = +0.482$; ΔAICc $= -13.5$) demonstrates that the drivers of alpaca density respond differently to climate and topography depending on where in Puno one looks. In the core puna, frost limitation dominates; in transition zones, market access and ecological factors take over. Spatially undifferentiated policies—a single insurance premium, a flat subsidy—would be systematically mis-targeted. Puno needs geographically tailored adaptation strategies, and the IVEA provides the spatial foundation on which to build them.

Future work should pursue three extensions. Kriging with External Drift, incorporating land cover or topographic wetness as covariates, could reduce the LOOCV error below the current 0.679 °C. A panel-data GWR model drawing on multiple census years would reveal whether the hotspot geography is stable or shifting under ongoing climate change [8]. And finally, expanding the IVEA to include adaptive-capacity indicators—household poverty, veterinary service access, and road density—would transform it from a climate-exposure index into a true climate-vulnerability index in the IPCC sense [10], which is the logical next step for this line of research.

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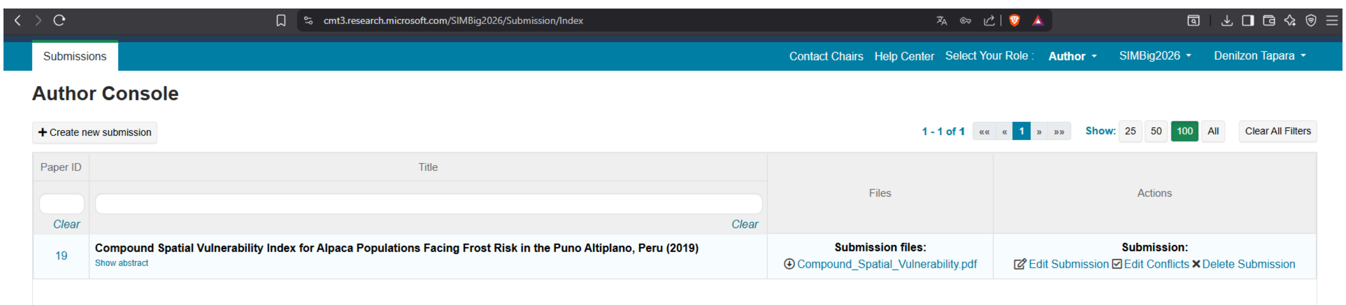

Figure 10. CAPTURA DE LA EVIDENCIA SUBMISION